

MATH 55 FINAL EXAM, PROF. SRIVASTAVA
DECEMBER 15, 2022, 11:40AM–2:30PM, 2050 VLSB.

Name: _____

SID: _____

GSI: _____

NAME OF THE STUDENT TO YOUR LEFT: _____

NAME OF THE STUDENT TO YOUR RIGHT: _____

INSTRUCTIONS: Write all answers clearly in the provided space. **Do not unstaple the exam.** Write your name and SID on every page. Write proofs in complete English sentences. To cite a theorem proved in class, just state it. **Be sure to fill in the multiple choice circles clearly.** Ambiguous multiple choice answers will be given a zero. Graph theory definitions are provided on the last page. You may use one 2–sided cheat sheet.

UC BERKELEY HONOR CODE: *As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others.*

Sign here: _____

Question	Points
1	12
2	12
3	12
4	12
5	12
6	12
7	14
8	14
Total:	100

Do not turn over this page until your instructor tells you to do so.
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1. Select true or false for each of the following. There is no need to provide an explanation.

- (a) (2 points) The propositions $\exists x(P(x) \vee Q(x))$ and $\exists xP(x) \vee \exists xQ(x)$ are logically equivalent.

☐ True ☐ False

- (b) (2 points) If n and k are positive integers with $k < n$, the function

$$f : \mathcal{P}(\{1, \dots, n\}) \rightarrow \mathcal{P}(\{1, \dots, k\})$$

defined by $f(S) = S \cap \{1, \dots, k\}$ is onto.

☐ True ☐ False

- (c) (2 points) If n and k are positive integers with $k < n$, the function

$$f : \mathcal{P}(\{1, \dots, n\}) \rightarrow \mathcal{P}(\{1, \dots, k\})$$

defined by $f(S) = S \cap \{1, \dots, k\}$ is one to one.

☐ True ☐ False

- (d) (2 points) The set

$$S = \{T : T \text{ is a tree with vertex set } \{1, \dots, n\} \text{ for some integer } n \geq 1\}$$

is uncountable.

☐ True ☐ False

- (e) (2 points) If n and r are positive integers with $r < n$ then

$$\binom{n+1}{r+1} = \binom{n}{r+1} + \binom{n}{r}.$$

☐ True ☐ False

- (f) (2 points) If X and Y are independent Geometric random variables both with parameter $q = 1/3$, then $X + Y$ is a Geometric random variable with parameter $q = 2/3$.

☐ True ☐ False

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2. Suppose that Nikhil is hungry, and selects a fruit by first picking one of two boxes labeled A and B uniformly at random, and then selecting a fruit uniformly randomly from that box. Box A contains 3 peaches, 3 apples, and 2 mangoes, and Box B contains 1 peach and 3 apples (and no mangoes).

(a) (6 points) Given that Nikhil selected a peach, what is the probability that he chose Box A ?

- ☐ $1/2$ ☐ $3/8$ ☐ $3/5$ ☐ $1/3$
☐ none of the above.

(b) (3 points) If Nikhil selected a mango, what is the probability that he chose Box B ?

- ☐ $1/2$ ☐ $3/12$ ☐ 0 ☐ 1
☐ none of the above.

(c) (3 points) Are the events “Nikhil selected Box A ” and “Nikhil selected a Peach” independent?

- ☐ Yes ☐ No

3. Suppose I throw 10 balls independently and uniformly at random into 5 bins labeled A, B, C, D, E . Call a bin *special* if it contains exactly 2 balls.

(a) (4 points) The expected number of special bins is:

- ☐ $\frac{10!}{32 \times 5^{10}}$ ☐ $\frac{5 \times 45 \times 4^8}{5^{10}}$ ☐ $\frac{5 \times 2 \times 9^8}{10^{10}}$ ☐ $\frac{5 \times 10 \times 4^8}{5^8}$
☐ none of the above.

(b) (3 points) The probability that every bin is special is:

- ☐ $(2/5)^5$ ☐ $(\frac{45 \times 4^8}{5^{10}})^5$ ☐ $\frac{10!}{32 \times 5^{10}}$ ☐ $1 - \frac{5 \times 120 \times 4^7}{5^{10}}$
☐ none of the above.

(c) (3 points) The probability that there exists a bin containing *at least two balls* is:

- ☐ $1 - \frac{5 \times 45 \times 4^8}{5^{10}}$ ☐ $2/5$ ☐ $\frac{5 \times 2^5}{5^5}$ ☐ 1
☐ none of the above.

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(d) (2 points) The events “Bin A is special” and “Bin B is special” are:

- ☐ independent ☐ not independent

4. The Math 555 final has 99 multiple choice questions numbered #1 – #99, each with 3 choices, *exactly one* of which is correct. The grading scheme is: every correct answer receives +1 point and every wrong answer receives $-1/2$ point. I have not studied at all, so I answer all of the questions independently and uniformly at random. Let X be the random variable equal to the total number of points I earn on the exam.

(a) (3 points) The expectation of X is:

- ☐ $99/3$ ☐ $1/3$ ☐ 0 ☐ $99/6$
☐ none of the above.

(b) (3 points) The variance of X is:

- ☐ $(99/3)^2$ ☐ $99/2$ ☐ $99(5/9)$ ☐ $99/6$.
☐ none of the above.

(c) (4 points) A *streak* is a sequence of 3 questions with adjacent numbers (e.g., 1,2,3 or 13,14,15) all of which I answered correctly. The expected number of streaks is:

- ☐ $\frac{99 \times 98 \times 97}{6} \times 1/27$ ☐ $97/27$ ☐ $97/3$ ☐ 33
☐ none of the above.

(d) (2 points) The events “I answer question #1 correctly” and “I answer question #99 correctly” are:

- ☐ independent ☐ not independent

5. (a) (4 points) A value of x for which $55x \equiv 5 \pmod{12}$ is:

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7
☐ 8 ☐ 9 ☐ 10 ☐ 11 ☐ none of the above.

(b) (4 points)] A value of y for which $54y \equiv 5 \pmod{12}$ is:

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7
☐ 8 ☐ 9 ☐ 10 ☐ 11 ☐ none of the above.

(c) (4 points) A value of z for which $z \equiv 54 \pmod{3}$ and $z \equiv 55 \pmod{4}$ is:

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7
☐ 8 ☐ 9 ☐ 10 ☐ 11 ☐ none of the above.

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6. (12 points) Prove or disprove: if x is a rational number and x^2 is an integer, then x must be an integer.

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7. A *dominarrow* is a 2×1 tile labeled with an arrow, which may be oriented in four ways as follows:



A *tiling* of a $2 \times n$ grid is a placement of dominarrows on the grid such that each square is occupied exactly once. Two tilings are the same iff they contain exactly the same dominarrows in the same locations with the same orientations.

- (a) (3 points) How many dominarrow tilings do the 2×1 grid and 2×2 grid have?
- (b) (8 points) Let a_n be the number of dominarrow tilings of the $2 \times n$ grid. Find a recurrence relation for a_n and prove your answer. Be sure to write down the base case(s).
- (c) (3 points) Using your answer to part (b), find a_4 .

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8. (a) (2 points) Give an example of a graph on 4 vertices with degree sequence 3, 2, 2, 1.

(b) (6 points) Suppose $G = (V, E)$ is a connected simple graph on n vertices with exactly n edges. Show that there is an edge $e \in E$ such that $G' = (V, E - \{e\})$ is a tree.

(c) (6 points) Suppose G is a connected simple graph on n vertices with exactly n edges. Show that G is 3-colorable.

Graph Theory Definitions

A **graph** is a pair $G = (V, E)$ where V is a finite set of **vertices** and E is a finite multiset of 2-element subsets of V , called **edges**. If E has repeated elements G is called a **multigraph**, otherwise it is called a **simple graph**. Two vertices $x, y \in V$ are **adjacent** if $\{x, y\} \in E$. An edge $e = \{x, y\}$ is said to be **incident with** x and y . The **degree** of a vertex x is the number of edges incident with it.

A **path** of length k in G is an alternating sequence of vertices and edges

$$x_0, e_1, x_1, e_2, \dots, x_{k-1}, e_k, x_k$$

where $e_i = \{x_{i-1}, x_i\} \in E$ for all $i = 1 \dots k$. The path is said to **connect** x_0, x_k , **visit** the vertices $x_0, \dots, x_k$, and **traverse** the edges $e_1, \dots, e_k$. A path is called **simple** if it contains no repeated vertices. If $x_0 = x_k$ then the path is called a **circuit**. A graph is **connected** if it contains a path between every pair of distinct vertices.

A **k-coloring** of G is a function $f : V \rightarrow \{1, \dots, k\}$ such that $f(x) \neq f(y)$ whenever x is adjacent to y . A graph is **k-colorable** if it has a k -coloring.

A graph $H = (W, F)$ is a **subgraph** of G if $W \subseteq V$ and $F \subseteq E$. If $W = V$ then H is called a **spanning** subgraph of G . A **connected component** of G is a maximal connected subgraph of G , i.e. a subgraph $H \subseteq G$ such that H is connected and adding any vertices or edges to H would make it disconnected.

A **cycle** is a circuit with no repeated edges and no repeated vertices, except the end-points. A graph is called **acyclic** if it does not contain a cycle as a subgraph. A connected acyclic graph is called a **tree**.

An **Eulerian circuit** of a connected multigraph G is a circuit which traverses every edge of G exactly once.